

W5 – Variational Inference (Part II)

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Last update: 19/8/18 8:57 pm

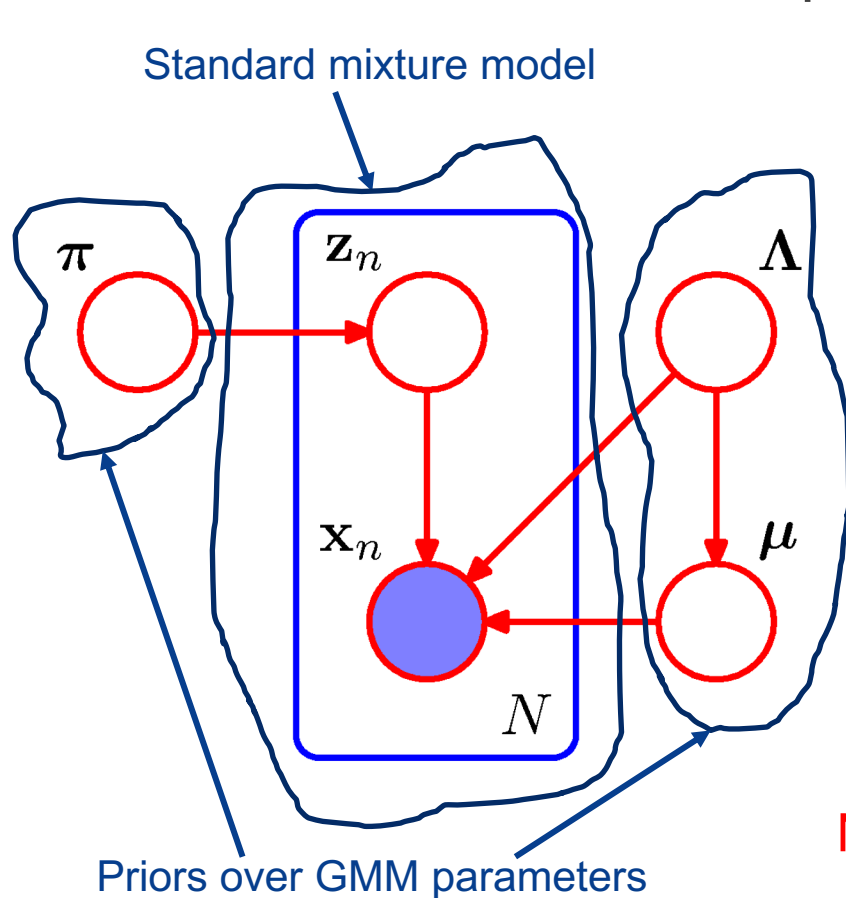
IV. Variational Inference in Bayesian GMMs

Bayesian GMMs (1)

$\mathbf{x}^{(n)}$: Observed variable for data point $n \rightarrow \mathbf{X} = \{\mathbf{x}^{(n)}\}$

$\mathbf{z}^{(n)}$: Hidden or missing variable $\rightarrow \mathbf{Z} = \{\mathbf{z}^{(n)}\}$

K : Number of mixture components



Prior over latent variables

$$p(\mathbf{Z}|\boldsymbol{\pi}) = \prod_{n=1}^N \prod_{k=1}^K \pi_k^{z_{nk}}$$

One-hot encoding

Conditional likelihood

$$p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = \prod_{n=1}^N \prod_{k=1}^K \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k^{-1})^{z_k^{(n)}}$$

Precision matrix

Need a prior over $\{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}\}$

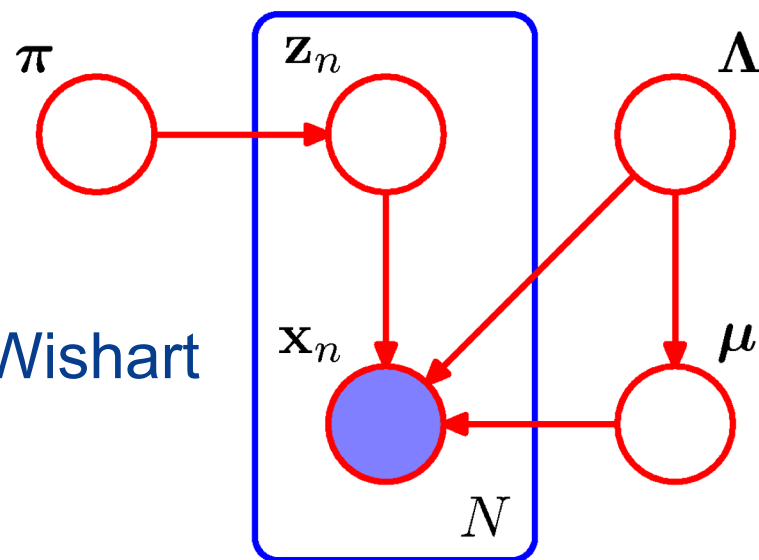
Bayesian GMMs (2)

- Dirichlet prior over mixture weights:

$$p(\boldsymbol{\pi}) = \text{Dir}(\boldsymbol{\pi}|\boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^K \pi_k^{\alpha_k - 1}$$

- Independent (Conjugate) Gaussian-Wishart prior over mean and precision:

$$p(\boldsymbol{\mu}, \boldsymbol{\Lambda}) = p(\boldsymbol{\mu}|\boldsymbol{\Lambda})p(\boldsymbol{\Lambda})$$



Brief Digression

The Gaussian-Wishart distribution

Assume:

$$\boldsymbol{\mu}_k \sim \mathcal{N}(\boldsymbol{\mu}_k | \tilde{\mathbf{m}}_k, (\tilde{\beta}_k \boldsymbol{\Lambda}_k)^{-1})$$

Wishart distribution (over PD matrices)

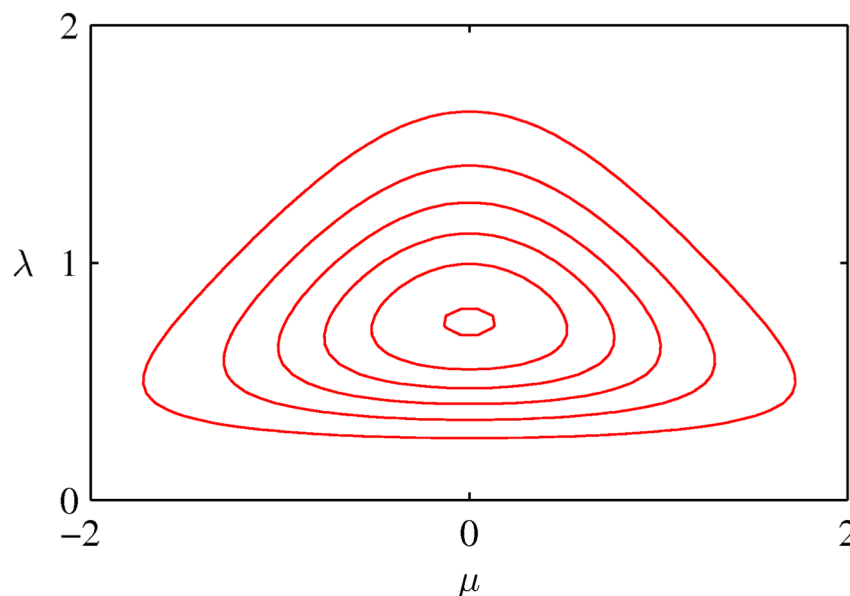
Generalization of the gamma distribution (to $D > 1$)

$$\boldsymbol{\Lambda}_k \sim \mathcal{W}(\boldsymbol{\Lambda}_k | \tilde{\mathbf{W}}_k, \tilde{\nu}_k)$$

$$\begin{aligned} \text{Then } p(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k) &= \mathcal{N}\mathcal{W}(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k | \tilde{\mathbf{m}}_k, \tilde{\beta}_k, \tilde{\mathbf{W}}_k, \tilde{\nu}_k) \\ &= \mathcal{N}(\boldsymbol{\mu}_k | \tilde{\mathbf{m}}_k, (\tilde{\beta}_k \boldsymbol{\Lambda}_k)^{-1}) \mathcal{W}(\boldsymbol{\Lambda}_k | \tilde{\mathbf{W}}_k, \tilde{\nu}_k) \end{aligned}$$

Gaussian-Wishart

- Joint distribution over mean and precision
- Generalization of the Gaussian-gamma distribution (to $D > 1$)



Bayesian GMMs (2)

- Dirichlet prior over mixture weights:

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- Independent (Conjugate) Gaussian-Wishart prior over mean and precision:

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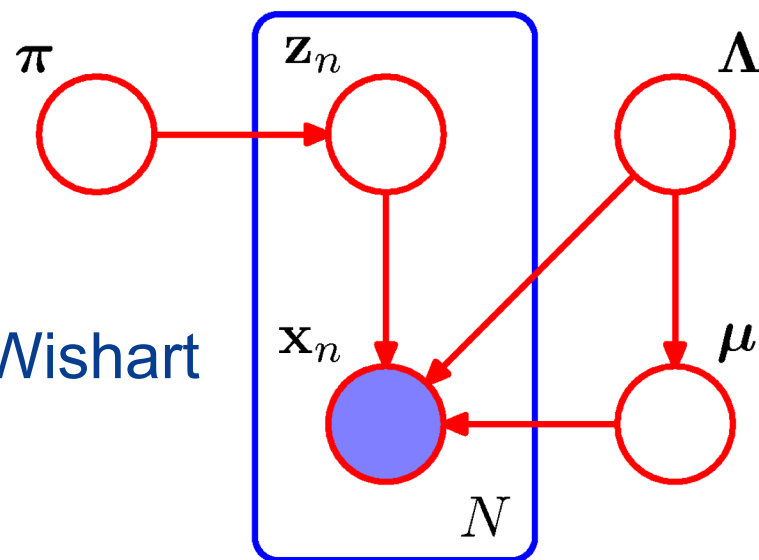
$$= \prod_{k=1}^K \mathcal{N}(\boldsymbol{\mu}_k | \mathbf{m}_0, (\beta_0 \boldsymbol{\Lambda}_k)^{-1}) \mathcal{W}(\boldsymbol{\Lambda}_k | \mathbf{W}_0, \nu_0)$$

- Full joint distribution:

$$p(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = p(\boldsymbol{\pi})p(\boldsymbol{\Lambda})p(\boldsymbol{\mu}|\boldsymbol{\Lambda})p(\mathbf{Z}|\boldsymbol{\pi})p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda})$$

– Note only $\mathbf{X}$ are observed

» Difference between $\mathbf{Z}$ and GMM parameters?



Variational Posterior Distribution (1)

- We need to define a posterior over all unobserved variables:

$$q(\mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = q(\mathbf{Z})q(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda})$$

- Note factorisation (independence) between $\mathbf{Z}$ and GMM parameters
- Remarkably, this is the only assumption we will make
 - » No constraints on the functional form
- Notational simplicity (subscripts on q and dependence on $\mathbf{X}$ dropped)
- Updates using general result for factorised distributions:

$$\log q^*(\mathbf{Z}) = \mathbb{E}_{\boldsymbol{\pi}}[\log p(\mathbf{Z}|\boldsymbol{\pi})] + \mathbb{E}_{\boldsymbol{\mu}, \boldsymbol{\Lambda}}[\log p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda})] + \text{const.}$$

– yielding:

$$q^*(\mathbf{Z}) = \prod_{n=1}^N \prod_{k=1}^K \tilde{r}_{nk}^{z_{nk}}$$

- $q^*(\mathbf{Z})$: same functional form as the prior
- $\tilde{r}_{nk}$: function of parameters of $q^*(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda})$
 - » Automatically satisfies the constraints
 - » Role of responsibilities as $\mathbb{E}_{z_{nk}}[\tilde{r}_{nk}]$

Variational Posterior Distribution (2)

- Expected sufficient statistics (analogous to EM statistics):

$$\tilde{r}_k = \sum_{n=1}^N \tilde{r}_{nk} \quad \tilde{\boldsymbol{\mu}}_k = \frac{1}{\tilde{r}_k} \sum_{n=1}^N \tilde{r}_{nk} \mathbf{x}^{(n)} \quad \tilde{\boldsymbol{\Sigma}}_k = \frac{1}{\tilde{r}_k} \sum_{n=1}^N \tilde{r}_{nk} (\mathbf{x}^{(n)} - \tilde{\boldsymbol{\mu}}_k)(\mathbf{x}^{(n)} - \tilde{\boldsymbol{\mu}}_k)^T$$

- For the variational posterior over GMM parameters:

$$q^*(\boldsymbol{\pi}) = \text{Dir}(\boldsymbol{\pi} | \tilde{\boldsymbol{\alpha}}) \quad - q^*(\boldsymbol{\pi}): \text{same functional form as the prior}$$
$$\tilde{\alpha}_k = \tilde{r}_k + \alpha_k$$

$$q^*(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k) = \mathcal{N}(\boldsymbol{\mu}_k | \tilde{\mathbf{m}}_k, (\tilde{\beta}_k \boldsymbol{\Lambda}_k)^{-1}) \mathcal{W}(\boldsymbol{\Lambda}_k | \tilde{\mathbf{W}}_k, \tilde{\nu}_k)$$

- $q^*(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k)$: same functional form as the prior

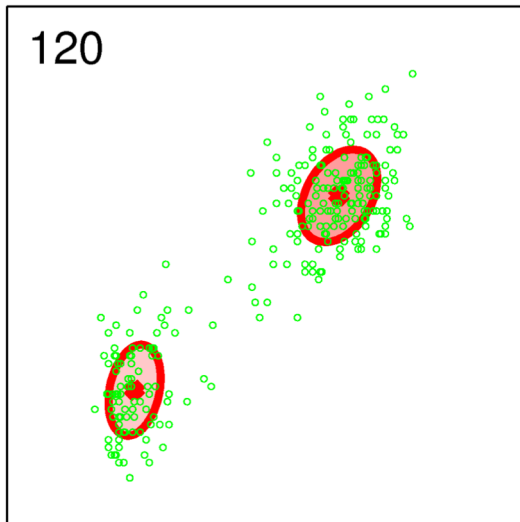
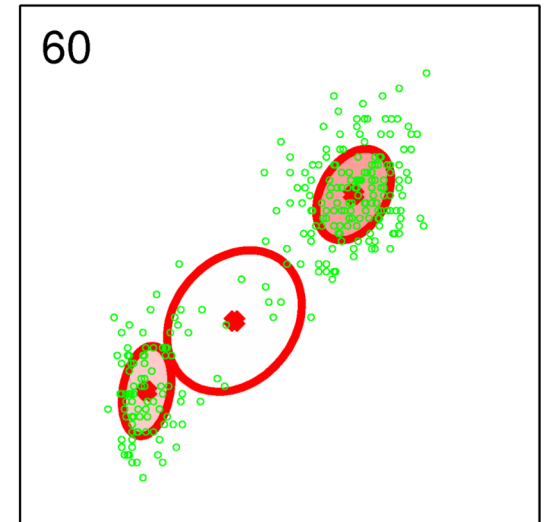
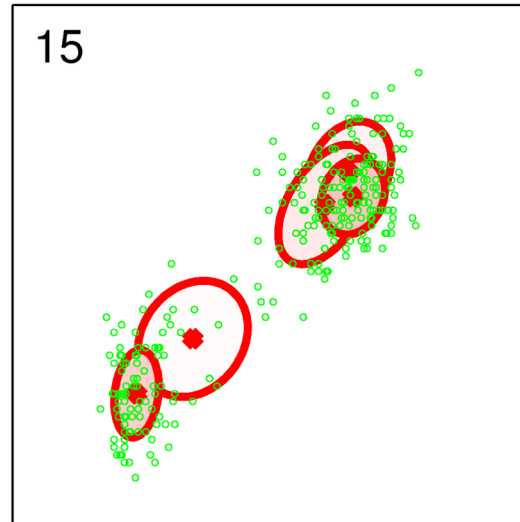
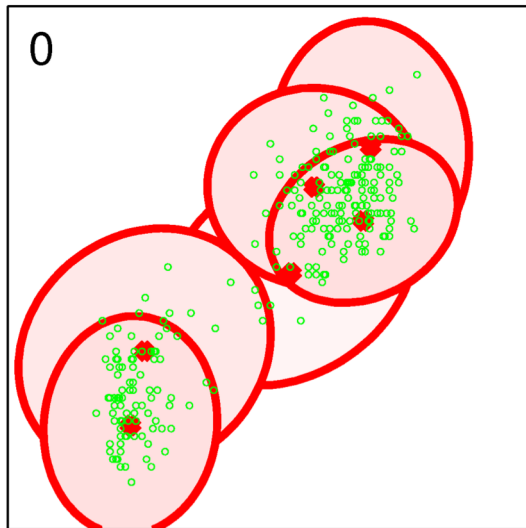
- $\tilde{m}_k, \tilde{\beta}_k, \tilde{\mathbf{W}}_k, \tilde{\nu}_k$: function of expected sufficient statistics (which include $\tilde{r}_{nk}$)

In fact, updates variational responsibilities $\tilde{r}_{nk}$, also analogous to EM's

Variational EM-like Algorithm

1. **VE-step:** Update variational responsibilities $\tilde{r}_{nk}$ Using current distributions over model parameters
2. **VM-step:** Update posterior over model parameters using current responsibilities
 - *Variational posteriors with same form as prior a consequence of conjugate priors*
 - *This is not the case for general priors or non-factorised distributions*
 - *Deterministic approximations using bounds*
 - *Assumptions on posterior functional form may be needed*
 - » *stochastic approximations of the variational objective*
 - » *Gradient-based optimization*

Variational Inference in Action

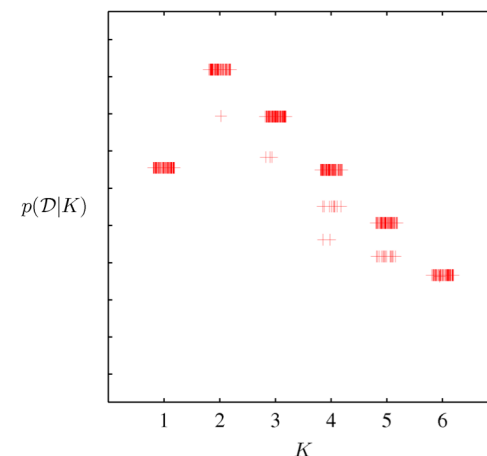


- Bayesian mixture with $K=6$
 - Ellipses: 1 std. dev. Contours
 - Density of red shade: mean of posterior mixing weights
 - Dirichlet prior to encourage sparsity: $\alpha=10^{-3}$
 - » Only two components with non-zero posterior mean weights

V. Final Remarks

Variational Inference: Remarks

- Close resemblance variational solution and EM's
 - As $N \rightarrow \infty \rightarrow$ ML solution by EM
 - Similar computational cost
 - Singularity problems do not appear in the Bayesian treatment
 - No overfitting
 - Can evaluate lower bound for converge assessment and debugging
- Alternative derivations
 - Write down lower bound and optimise wrt posterior parameters
 - » Exploit knowledge about conjugacy
- Determining number of components
 - Model selection needs to consider $K!$ settings
 - » Adding penalty term $\log K!$ to the lower bound
 - » Alternative to cross-validation



Advantages and Disadvantages of VI

(Slide based on Shakir Mohamed's tutorial on VI)

Disadvantages

- Never exact, only approximate posterior
- Typically underestimate the variance
- Can get stuck in local optima
- Limited theory

Advantages

- Applicable to a large class of models
- Can assess convergence
- Can do model selection
- Usually scalable and fast
- Compact representation of posterior

Conclusions

- Variational inference (VI) as a deterministic approximate algorithm for posterior estimation
 - Integration problem transformed into an optimisation problem
 - Optimisation of the variational objective (evidence lower bound) equivalent to minimizing $KL(q||p)$
 - usually fast and scalable but inexact
- Possible solutions
 - Free-form posterior
 - » Factorised distributions and mean-field solution
 - » Closed-form updates through conjugacy
 - Parameterized posterior
- Reading
 - Bishop (PRML, 2006): Ch. 10 (except Sec. 10.4, 10.5, 10.7), Sec 2.3.6
 - Murphy (MLaPP, 2012): Ch 21 (except Sec. 21.4, 21.7, 21.8)